

Object Oriented Programming
(OOPS)
LAB FILE

SEMESTER:- 4

SUBMITTED BY:- HARSHIT KUMAR

500122407

B-1

SUBMITTED TO:- MR. SATYAM TIWARI SIR

EXPERIMENT-1

TITLE: Introduction to Java Environment

- 1. Explore and understand the role of JDK, JRE and JVM.**
- 2. Install latest available JDK and verify the Java Environment.**
- 3. Create a Sample Hello World Program using simple text editor (e.g. Notepad) and show the steps to compile and execute the program using command prompt.**
- 4. Display your name and complete address in different lines.**

1. Explore and Understand the Role of JDK, JRE, and JVM

JDK (Java Development Kit):

1. A software development kit used to write and compile Java programs.
2. Includes tools like javac (compiler), java (runtime), and libraries for Java development.

JRE (Java Runtime Environment):

1. Provides the libraries, Java Virtual Machine (JVM), and other components to run Java applications.
2. It's a subset of the JDK.

JVM (Java Virtual Machine):

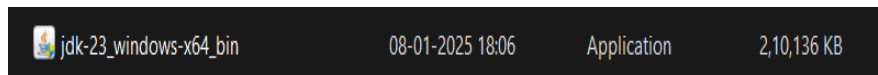
1. Runs Java bytecode on your machine.
2. Converts the bytecode (compiled Java code) into machine code for execution.
3. Ensures platform independence.

2. Install Latest Available JDK and Verify the Java Environment

Download and Install JDK:

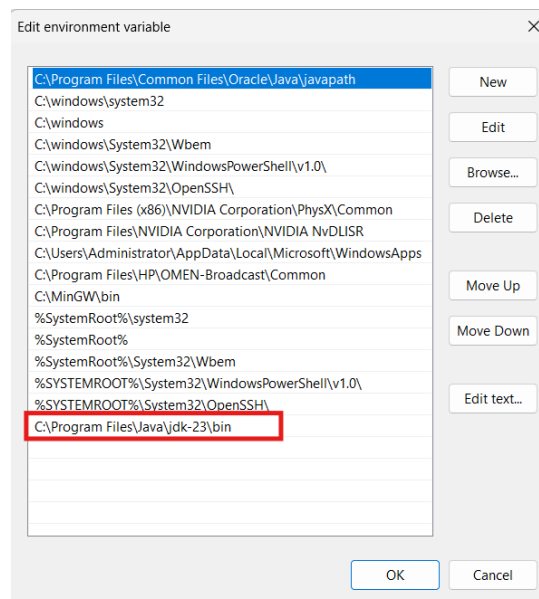
Visit [Oracle JDK website](#) or OpenJDK and download the latest version of JDK for your OS.

Install it by following the instructions provided.



Set Up Environment Variables:

Add the JDK bin directory to your system's PATH.



On Windows:

Right-click This PC > Properties > Advanced system settings > Environment Variables.

Add C:\Program Files\Java\jdk<version>\bin to the Path variable.

Verify Installation:

Open a terminal or command prompt and type:

java -version

The output should show the installed JDK version.

```
C:\Users\HARSHIT>java -version
java version "23.0.1" 2024-10-15
Java(TM) SE Runtime Environment (build 23.0.1+11-39)
Java HotSpot(TM) 64-Bit Server VM (build 23.0.1+11-39, mixed mode, sharing)
```

3. Create a Sample Hello World Program and Execute It

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

```
Microsoft Windows [Version 10.0.26100.2894]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HARSHIT>D:

D:\>cd JavaProjects

D:\JavaProjects>javac HelloWorld.java

D:\JavaProjects>java HelloWorld
Hello, World!

D:\JavaProjects>
```

4. Display Your Name and Address in Different Lines

```
public class Address{  
    public static void main(String[] args) {  
        System.out.println("Harshit Kumar");  
        System.out.println("P-208");  
        System.out.println("Shivalik Nagar Phase 3");  
        System.out.println("Haridwar, Uttarakhand, 249403");  
    }  
}
```

```
Microsoft Windows [Version 10.0.26100.2894]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\HARSHIT>d:  
  
D:\>cd JavaProjects  
  
D:\JavaProjects>javac Address.java  
  
D:\JavaProjects>java Address  
Harshit Kumar  
P-208  
Shivalik Nagar Phase 3  
Haridwar, Uttarakhand, 249403  
  
D:\JavaProjects>
```